

Notes: Antill (JFE, 2022)
Do the Right Firms Survive Bankruptcy?

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1 Big picture

- **Question.** Do Chapter 11 exit choices maximize creditor recovery, or do inefficient liquidations and acquisitions occur?
- **Forms of exit.** A bankrupt firm may reorganize, be acquired as a going concern, or be liquidated piecemeal.
- **Challenge.** The recovery under the unchosen exit form is unobserved, and participants choose exit forms using private recovery signals.
- **Strategy.** The paper uses within-district random judge assignment and judge liquidation taste to estimate a generalized Roy model.
- **Main finding.** Inefficient piecemeal liquidations are frequent and costly, especially when implemented through 363 sales.
- **Policy message.** Better screening of 363 sales could improve creditor recovery more effectively than a blanket ban.

2 Data and measurement

The sample contains 503 large public Chapter 11 bankruptcies. Recovery rates come from Moody's Ultimate Recovery Database. Bankruptcy characteristics and 363-sale classifications come from BankruptcyData.com. The main outcome is family recovery: total payments to all debtholders, discounted to the bankruptcy filing date, divided by total prepetition debt.

Exit outcomes are reorganization, liquidation, and acquisition. A Section 363 sale is an asset sale outside the plan-confirmation voting process. Judge liquidation taste measures a judge's historical tendency to push cases toward liquidation and is used as an instrument.

3 Models and empirical specifications

3.1 Potential recoveries

$$Y_i(1) = \text{recovery if reorganization}, \quad Y_i(0) = \text{recovery if non-reorganization.}$$
$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

3.2 IV regression

$$Y_i = \alpha + \tau D_i + X_i \beta + \varepsilon_i, \quad D_i = \pi_0 + \pi_1 \text{JudgeLiquidationTaste}_i + X_i \pi + \nu_i.$$

3.3 Generalized Roy model

$$Y_i(1) = X_i \beta_1 + U_{1i}, \quad Y_i(0) = X_i \beta_0 + U_{0i}.$$

The model accounts for selection on unobserved recovery signals.

3.4 Marginal treatment effect

$$\Delta_{MTE}(x, p) = x(\beta_1 - \beta_0) + K'(p).$$

3.5 Inefficiency losses

$$LR_i = 1(ATT(X_i) < 0)|ATT(X_i)|D_i,$$

$$LL_i = 1(ATUT(X_i) > 0)ATUT(X_i)1(Liquidation_i),$$

$$LA_i = 1(ATUT(X_i) > 0)ATUT(X_i)1(Acquisition_i).$$

4 Identification and empirical design

The design uses within-district random assignment of bankruptcy judges. Liquidation-inclined judges reduce the probability of reorganization and increase the use of 363 sales. The key exclusion restriction is that judge liquidation taste affects recovery only through the form of exit. Figure 1 supports propensity-score overlap; appendix exercises support monotonicity and model fit. IV identifies local effects for compliers, while the structural Roy model extrapolates to counterfactual policy rules.

5 Tables and figures

5.1 Table 1: Summary Statistics

Read: The sample has 503 cases, mean family recovery of 0.49, reorganization share of 0.66, liquidation share of 0.25, acquisition share of 0.10, and 363-sale share of about 0.51.

Detailed interpretation: The sample is large and economically important. The high 363-sale frequency makes the institutional channel central.

Economic message: Exit-form choices have large consequences for creditor recoveries.

Table 1
Summary statistics.

This table shows the summary statistics. *Reorganization*, *Liquidation*, and *Acquisition* are indicators for mutually exclusive forms of exit, as categorized by Moody's Ultimate Recovery database (URD). The binary variable *363 Sale* is an indicator for asset sales under Section 363(b), as categorized by New Generation Research's Bankruptcydata.com (BD). *Family Recovery* (from the URD) is the ratio of the total dollar amount paid out to all the debtholders at the end of the bankruptcy, discounted to the start of the bankruptcy, to the total dollar amount of pre-petition debt. *Leverage* is the ratio of the total dollar amount of pre-petition debt (from the URD) to the total book assets at the time of filing (from BD). *Junior Debt Share* is the ratio of the total dollar amount of junior debt (as classified by the URD) to the sum of junior and senior debt (from the URD). *Secured Debt Share* is defined analogously. The *Baa-Treasury Spread* is the spread between Moody's seasoned-Baa-corporate-bond yield and the yield on a 10-year-constant-maturity Treasury, measured in the month of the bankruptcy filing. *Judge Experience* is the number of months between the bankruptcy judge's appointment date (from the LoPucki bankruptcy database) and the date of the filing. The other variables, which are measured at the time of filing from BD, are self-explanatory.

	Mean	SD	P10	P90	N
<i>Family Recovery</i>	0.49	0.29	0.12	0.98	503
<i>Assets, ln (2018 dollars)</i>	3.08	9.51	0.23	5.34	503
<i>Employees, thousands</i>	9.00	19.20	0.37	21.10	503
<i>Leverage</i>	0.73	0.53	0.25	1.22	503
<i>Revenue/Assets</i>	1.19	0.96	0.22	2.46	503
<i>Judge Experience, Months</i>	118.12	87.49	24.00	230.00	503
<i>Junior Debt Share</i>	0.48	0.32	0.00	0.90	503
<i>Secured Debt Share</i>	0.54	0.35	0.00	1.00	503
<i>Baa-Treasury Spread</i>	2.64	0.98	1.67	3.53	503
<i>Reorganization</i>	0.66	0.47	0.00	1.00	503
<i>Liquidation</i>	0.25	0.43	0.00	1.00	503
<i>Acquisition</i>	0.10	0.29	0.00	0.00	503
<i>363 Sale</i>	0.51	0.50	0.00	1.00	497

Table 2
First-stage estimates.

This table shows results of ordinary least squares (OLS) regressions in which the dependent variable is (i) an indicator for reorganization or (ii) an indicator for 363 sales. I calculate the judge-liquidation-taste instrument, using New Generation Research's Bankruptcydata.com (BD), as the fraction of cases that the bankruptcy judge converts to Chapter 7 in a test sample of bankruptcies that is disjoint from my main sample. See the text for details. All regressions include industry fixed effects, defined using 2-digit SIC codes from the Moody's Ultimate Recovery database. The other listed controls are described in Table 1. All regressions include filing district fixed effects and filing district-specific time trends for the filing year, both from BD. Standard errors, clustered at the filing district level, are presented in parentheses. In the regression for column 1 I define acquisitions as liquidations. In the regression for column 2, I define acquisitions as reorganizations. In the regression for column 3, I exclude acquisitions, which accounts for the sample size of 455. In the regression for column 4, I use an indicator for 363 sales, which is missing for six observations, as the dependent variable. *, **, and *** denote 10, 5, and 1% statistical significance, respectively.

	Reorganization			363 Sale
	(1)	(2)	(3)	(4)
<i>Liquidation Taste</i>	-0.419*** (0.126)	-0.403*** (0.113)	-0.463*** (0.125)	0.420*** (0.130)
<i>Leverage</i>	0.094 (0.074)	0.092 (0.064)	0.101 (0.075)	-0.041 (0.044)
<i>Log(Assets)</i>	0.091*** (0.026)	0.079*** (0.031)	0.089*** (0.031)	0.100*** (0.020)
<i>Revenue/Assets</i>	-0.058 (0.038)	-0.063*** (0.020)	-0.076*** (0.034)	-0.013 (0.028)
<i>Log(Employees)</i>	-0.017 (0.023)	-0.019 (0.025)	-0.012 (0.024)	0.022 (0.019)
<i>Baa-Treasury Spread</i>	-0.033 (0.032)	-0.014 (0.019)	-0.016 (0.027)	-0.050* (0.026)
<i>Junior Debt Share</i>	0.075 (0.065)	0.086 (0.066)	0.102* (0.057)	0.106 (0.065)
<i>Secured Debt Share</i>	0.014 (0.113)	0.062 (0.092)	0.062 (0.095)	0.039 (0.085)
<i>Log(Judge Experience)</i>	0.058*** (0.012)	0.052*** (0.023)	0.052*** (0.018)	-0.031 (0.042)
<i>Industry FE</i>	Y	Y	Y	Y
<i>Court FE and Trends</i>	Y	Y	Y	Y
<i>Acquisitions</i>	Liquidation	Reorganization	Excluded	N/A
<i>Instrument F-Statistic</i>	10.61	12.69	13.79	10.44
<i>Observations</i>	503	503	455	497
<i>Adj. R²</i>	0.0919	0.0812	0.0982	0.137

es; (iii) the logarithm of the book value of the firm's assets (in 2018 dollars); (iv) the firm's leverage; (v) the fraction of the firm's total debt that is junior to other debt as defined by the URD; (vi) the fraction of the firm's total debt that is secured (as defined by the URD); (vii) the spread between Moody's Baa-corporate-bond yield and the 10-year-constant-maturity-Treasury yield in the month of filing; and (viii) judge experience, a proxy for judge skill, which I calculate as the logarithm of the number of months since the judge was appointed (Iversen et al.,

5.2 Table 2: First-Stage Estimates

Read: Judge liquidation taste lowers reorganization probabilities and raises 363-sale probabilities.

Detailed interpretation: This table validates the judge-taste instrument.

Economic message: Judge assignment shifts the bankruptcy outcome.

5.3 Table 3: Instrumental Variables Regressions

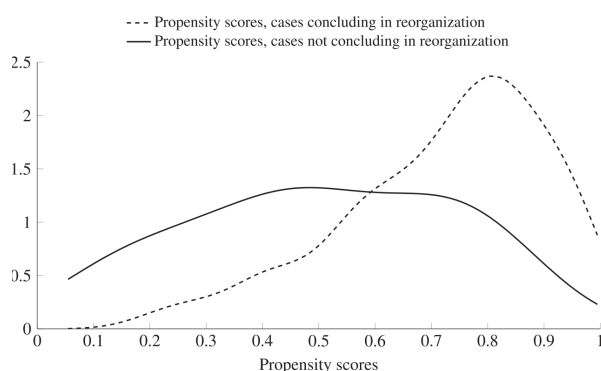
Read: Reorganization raises recovery for compliers; 363 sales reduce recovery.

Detailed interpretation: The table provides reduced-form causal evidence motivating the structural model.

Economic message: Reorganization is recovery-enhancing for judge-marginal cases.

Table 3
Instrumental variables regressions.
This table shows results of 2SLS regressions of family recovery on indicators for either reorganization or 363 sales. See Tables 1 and 2 for variable definitions. The text provides further details. In columns 1–3, I show results of 2SLS regressions of family recovery on an indicator for reorganization, instrumented by judge liquidation taste. In the regression for column 1, I define acquisitions as liquidations. In the regression for column 2, I define acquisitions as reorganizations. In the regression for column 3, I exclude acquisitions, which accounts for the sample size of 455. In the regression for column 4, I use an indicator for 363 sales, which is missing for six observations, as the endogenous variable. In addition to the listed controls, all regressions include industry fixed effects, filing district fixed effects, and district-specific time trends for the filing year. Standard errors, clustered at the district level, are presented in parentheses. *, **, and *** denote 10, 5, and 1% statistical significance, respectively.

	Family recovery			
	(1)	(2)	(3)	(4)
Reorganization	0.534** (0.244)	0.534* (0.315)	0.423* (0.221)	
363 Sale				-0.551** (0.230)
Leverage	-0.145*** (0.037)	-0.145*** (0.035)	-0.150*** (0.036)	-0.114*** (0.025)
Log(Assets)	-0.016 (0.031)	-0.010 (0.028)	-0.010 (0.024)	0.087*** (0.030)
Revenue/Assets	0.053*** (0.017)	0.056*** (0.022)	0.047** (0.019)	0.014 (0.014)
Log(Employees)	-0.016 (0.013)	-0.015 (0.011)	-0.020* (0.012)	-0.012 (0.018)
Baa-Treasury Spread	-0.002 (0.019)	-0.012 (0.013)	-0.006 (0.015)	-0.050*** (0.013)
Junior Debt Share	-0.201*** (0.057)	-0.208*** (0.057)	-0.211*** (0.053)	-0.107*** (0.038)
Secured Debt Share	0.160*** (0.040)	0.135** (0.054)	0.144*** (0.050)	0.183*** (0.039)
Log(Judge Experience)	-0.033*** (0.011)	-0.030 (0.022)	-0.022* (0.012)	-0.021 (0.019)
Industry FE	Y	Y	Y	Y
Court FE and Trends	Y	Y	Y	Y



5.4 Figure 1: Propensity Scores

Read: Reorganization and non-reorganization cases have overlapping propensity-score distributions.

Detailed interpretation: Overlap supports the Roy model's counterfactual comparisons.

Economic message: The data contain enough common support for model-based decision rules.

5.5 Table 4: Counterfactuals

Read: Inefficient liquidations reduce recovery by 0.10 per dollar of debt claim, inefficient acquisitions by 0.04, and inefficient reorganizations by only 0.02. The achievable gain is 0.12.

Detailed interpretation: The main value loss is from too many liquidations, not too many reorganizations.

Economic message: Model-guided exit selection could substantially improve creditor recovery.

5.6 Figure 2: Decomposition of Creditor Losses

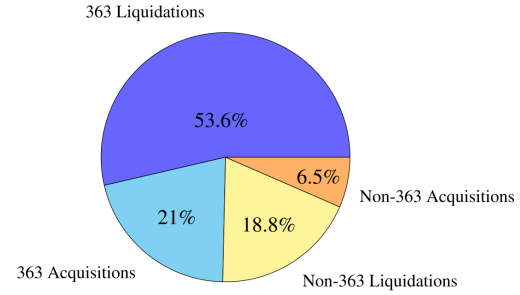
Read: Piecemeal liquidation involving 363 sales accounts for roughly 54% of inefficient non-reorganization losses; 363-sale outcomes together account for about 75%.

Table 4

Counterfactuals.

This table shows results of counterfactual exercises from the generalized Roy model. Details of the generalized Roy model estimation are provided in [Appendix C.1](#). Column 1 displays estimates of my counterfactuals. I bootstrap the entire model estimation and the calculation of each counterfactual 500 times, clustering at the district level. Column 2 shows the fraction of bootstrapped estimates that are strictly negative. See [Section 5.4](#) for detailed definitions of the counterfactuals involving 363 sales. See [Section 5.3](#) for detailed definitions of the other counterfactuals.

	Estimate (1)	$F(0)$ (2)
<i>Recovery-rate changes</i>		
No inefficient reorganizations	0.02	0.00
No inefficient liquidations	0.10	0.00
No inefficient acquisitions	0.04	0.00
Achievable gain	0.12	0.04
Achievable gain, OOS	0.11	0.06
Achievable gain in cases involving 363 sales	0.09	0.03
No liquidations involving 363 sales	0.03	0.23
No acquisitions involving 363 sales	0.01	0.26
No inefficient acquisitions or liquidations involving 363 sales	0.10	0.00
<i>Unconditional frequencies</i>		
Inefficient liquidations	0.15	0.00
Inefficient acquisitions	0.06	0.00
Inefficient reorganizations	0.04	0.00
<i>Regression of inefficiency on an indicator for 363 sales</i>		
Regression coefficient on 363 sales indicator	0.25	0.01



Detailed interpretation: Losses are concentrated in 363-sale liquidations.

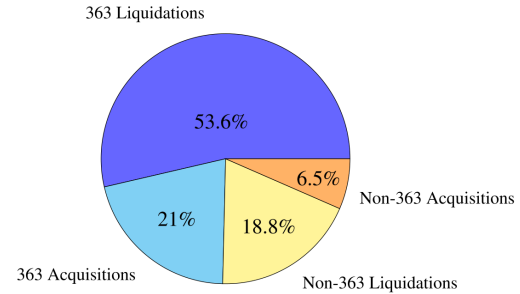
Economic message: The 363-sale process is the main institutional source of inefficiency.

Table 4

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	Estimate (1)	$F(0)$ (2)
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No inefficient reorganizations	0.02	0.00
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Achievable gain	0.12	0.04
Achievable gain, OOS	0.11	0.06
Achievable gain in cases involving 363 sales	0.09	0.03
No liquidations involving 363 sales	0.03	0.23
No acquisitions involving 363 sales	0.01	0.26
No inefficient acquisitions or liquidations involving 363 sales	0.10	0.00
<i>Unconditional frequencies</i>		
Inefficient liquidations	0.15	0.00
Inefficient acquisitions	0.06	0.00
Inefficient reorganizations	0.04	0.00
<i>Regression of inefficiency on an indicator for 363 sales</i>		
Regression coefficient on 363 sales indicator	0.25	0.01



5.7 Figure 3: Decomposition of the Achievable Gain

Read: Preventing inefficient piecemeal liquidations involving 363 sales accounts for about 56% of the achievable gain; 363-sale non-reorganization outcomes account for about 77%.

Detailed interpretation: A targeted 363-sale screening rule would capture most of the improvement.

Economic message: Targeted policy beats a blanket ban.

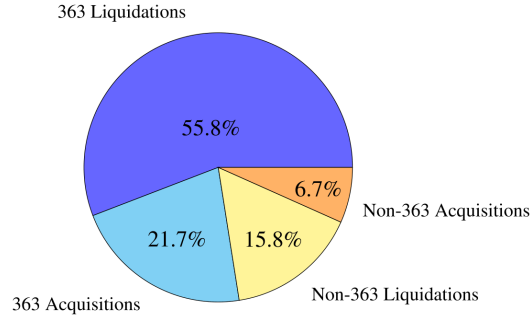


Figure 3: Decomposition of the achievable gain

6 Short summary

- The paper studies whether Chapter 11 exit choices maximize creditor recovery.
- Judge liquidation taste provides exogenous variation in exit form.
- IV estimates show reorganization raises recovery for compliers.
- The Roy model estimates counterfactual recoveries and feasible policy gains.
- Inefficient piecemeal liquidations are the largest source of creditor losses.
- Losses and gains are concentrated in 363-sale cases.

7 Conclusion

7.1 Core conclusion

Piecemeal liquidations often reduce creditor recovery relative to reorganization. Acquisitions are less harmful, and inefficient reorganizations are relatively rare.

7.2 Economic intuition

The main problem is destruction of going-concern value. 363 sales can bypass creditor approval and allow quick asset sales even when reorganization would maximize recovery.

7.3 Policy implication

The paper supports stricter screening of 363 sales or model-guided recommendations, rather than a blanket ban on all 363 sales.

7.4 Limitations

The analysis focuses on creditor recovery, relies on judge-taste instruments and structural modeling, and does not quantify delay costs or broader social welfare.

7.5 Takeaway

The bankruptcy system often uses private information well through plan confirmation, but the 363-sale process creates a concentrated source of inefficient liquidation.